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Task-Specific Manipulability Metrics for Redundancy Optimization in Cooperative Manipulation

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Abstract

Purpose – This paper investigates how redundant degrees of freedom can improve task-relevant dexterity and force capability in cooperative manipulation without disturbing commanded object motion. It compares task-specific manipulability metrics, particularly when the dominant external-load direction differs from the commanded motion direction.

Design/methodology/approach – A closed-loop inverse-kinematics and null-space gradient-projection framework is used to compare velocity, force, and directional-force manipulability. The objectives are evaluated on a planar dual-arm hardware platform using external loads and actuator-current-based effort measurement. A complementary MuJoCo study with two 7-DoF Franka Panda manipulators extends the comparison to static holding, translational pick-and-place, and six-dimensional pick-and-place from six initial object configurations.

Findings – Hardware experiments show that the objectives produce distinct internal configurations while preserving object motion. Velocity manipulability improves kinematic dexterity, while directional-force manipulability yields lower effort than force-only optimization when the commanded motion and external-load directions differ. The spatial simulations further show that all three objectives improve or maintain their respective measures across different initial configurations while maintaining low object-tracking errors.

Originality/value – The paper provides an implementation-oriented comparison of velocity, force, and directional-force manipulability objectives and offers practical guidance for selecting redundancy objectives according to task and external-load direction.

Keywords Dual robotic arms, redundant manipulators, cooperative manipulation, bimanual manipulation, manipulability, null-space control

Paper Type Research paper

1. Introduction

Cooperative manipulation refers to the physical handling of a common object by two or more robotic arms, where the collective effort enables tasks that exceed the capabilities of individual manipulators [Caccavale and Uchiyama \(2008\)](#); [Abbas et al. \(2023\)](#). Such tasks include carrying large payloads, assembling multiple parts without special fixtures, and performing dexterous manipulation, such as multi-fingered grasping, where coordinated finger motions enable stable, adaptive, and precise control of complex objects [Caccavale and Uchiyama \(2008\)](#); [Khatib \(1986\)](#). In teleoperated systems, effective redundancy resolution is crucial as it ensures that the operator’s bimanual dexterity is faithfully transmitted to

the remote manipulators, enabling intuitive and robust control even under environmental disturbances and kinematic constraints [Mason and Salisbury Jr \(1985\)](#). Figure 1 illustrates this redundancy through different feasible joint configurations for the same grasped-object pose.

With recent advancements in humanoid robotics and the increasing research focus on anthropomorphic multi-fingered hands and bi-manual manipulation, there is a growing demand for more dexterous and human-like robotic capabilities [Tong et al. \(2024\)](#); [Billard and Kragic \(2019\)](#); [Grannen et al. \(2023\)](#); [Abbas et al. \(2023\)](#). In such humanoid-oriented applications, redundancy enables collision-free, energy-efficient, and dynamically stable cooperative motions.

Existing approaches to redundancy optimization have

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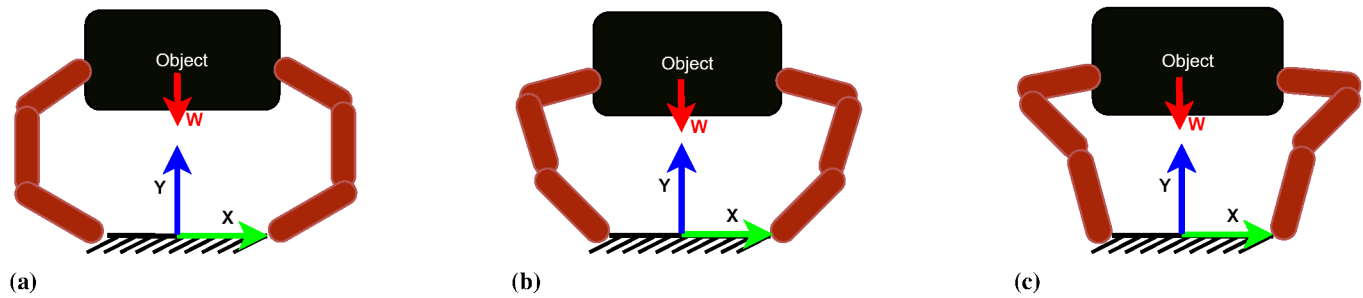


Figure 1: Different joint configuration solutions for a 3-DOF planar dual-arm manipulator grasping an object. Each subfigure shows a distinct inverse kinematics solution that maintains contact with the object while varying joint angles to leverage redundancy for enhanced performance. **Source:** Authors' own work.

primarily focused on single-arm manipulators using methods such as null-space projection, task-priority control, and weighted redundancy resolution to optimize degrees of freedom (DOF) for obstacle avoidance and energy efficiency Nakamura (1991); Freddi et al. (2016); Yoshikawa (1985); Nakamura et al. (1987); Nakamura and Hanafusa (1987); Chiaverini (1997). In contrast, recent works address cooperative dual-arm or multi-fingered systems by formulating the inverse kinematics problem as a global optimization task. For example, the method in Hernández-Barragán et al. (2019) employs a differential evolution algorithm to solve dual-arm inverse kinematics while avoiding Jacobian inversion and handling joint limits via penalty functions; however, it operates offline and may not respond to disturbances. Similarly, the approach in Geng et al. (2024) combines global planning with a local QP-based reactive module to achieve collision avoidance and coordinated motion in confined environments, yet it relies on heuristic global planning and suffers from a limited planning horizon.

Beyond scalar volume maximization, manipulability has also been formulated in terms of compatibility between robot posture and task-specific motion or force requirements Chiu (1988); Lee (1989). For cooperative systems, object-level velocity and force manipulability have been extended to multiple-arm settings, including absolute and relative motion and external and internal force components Chiacchio et al. (1991). More recent work has used directional manipulability to select configurations suited to task-dependent motion, force generation, and tool-use requirements Ti et al. (2024), while force- and stiffness-feasibility studies have shown that redundant posture selection influences directional interaction capability and actuator saturation Ajoudani et al. (2017). The present work adopts this task-compatible viewpoint and compares velocity, force, and directional-force objectives at the grasped-object level, particularly when commanded motion and external-load directions are not aligned.

Unlike approaches that emphasize global planning, heuristic search, or offline redundancy resolution along prescribed paths Zhou and Nguyen (1997); Hayashi and Kuipers (1991); Gal and Zarrouk (2022); Yin et al. (2024), this work uses an established closed-loop inverse-kinematics and null-space

gradient-projection structure as a common implementation basis. The objective is not to introduce a new redundancy-resolution algorithm, but to evaluate how different task-specific manipulability metrics influence cooperative manipulation performance under the same real-time control pipeline. This framing allows velocity manipulability, force manipulability, and directional-force manipulability to be compared in terms of posture adaptation, preservation of the commanded object motion, and actuator-current effort under directional external loading.

The framework supports task-specific selection of manipulability objectives, allowing users to optimize for either velocity or force depending on operational needs. In a *velocity-optimization mode*, for instance, the system maximizes its ability to move the object in arbitrary directions, improving dexterity for tasks such as fine alignment or orientation adjustment. Conversely, in a *force-optimization mode*, the method effectively distributes loads across joints, helping to prevent actuator saturation and reduce motor stress during high-load tasks. These capabilities make the proposed method versatile and applicable to a broad class of cooperative manipulation problems, including those encountered in humanoid robotics, teleoperation, and industrial assembly. A qualitative example of this effect is shown in Figure 2, where redundancy optimization leads to more favorable configurations than the corresponding non-optimized execution.

In summary, the key contributions include:

- An implementation-oriented cooperative manipulability-shaping pipeline that combines grasp compatibility, hand Jacobian kinematics, and null-space redundancy resolution within a common control structure for dual-arm and multi-contact manipulation.
- A direction-aware force manipulability objective for tasks where the dominant external load direction differs from the desired object motion direction, such as lifting, constrained pushing, insertion, and tool-use operations.
- A comparative evaluation of the direction-aware force objective against two standard redundancy objectives, namely velocity manipulability and force manipulability. This comparison clarifies how different objective choices

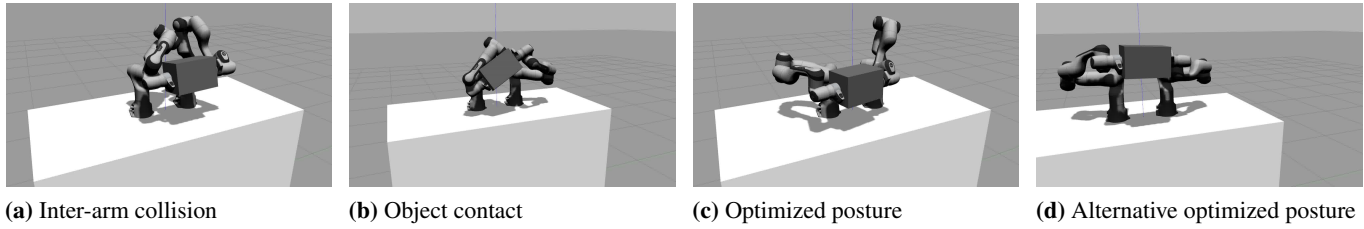


Figure 2: Comparison of dual-arm manipulation with and without velocity manipulability optimization. Without optimization, the execution can lead to inter-arm collision or unintended contact between a manipulator link and the object. With velocity manipulability optimization, the controller reaches configurations that avoid the illustrated collision/contact cases in this experiment while improving manipulability. **Source:** Authors' own work.

reshape the internal joint configuration while preserving the commanded object motion.

- Hardware validation on a planar dual-arm manipulation platform showing that the most suitable redundancy objective depends on the task condition and external load direction. In particular, the experiments compare velocity, force, and directional-force objectives in terms of manipulability improvement, posture adaptation, and actuator-current effort.
- A complementary 7-DoF dual-Franka simulation study that extends the comparison of velocity, force, and directional-force manipulability objectives to spatial bi-manual manipulation across static holding, translational pick-and-place, and full six-dimensional pick-and-place tasks from multiple initial configurations.

Supplementary Video 1. Representative planar dual-arm hardware experiments for the baseline and optimized executions, including velocity, force, and directional-force manipulability cases, together with the spatial dual-Franka simulation study, are available at: <https://youtu.be/CubFLF5DAzE>.

2. Mathematical Formulation

2.1 Grasp Theory Background

In grasp theory, the *grasp matrix* G maps the forces (and possible moments) applied at the contact points to the resultant wrench (force-torque vector) on the object, while the *hand Jacobian* J_H transforms joint velocities into contact frame velocities (Babin and Gosselin (2021)).

2.1.1 Grasp Matrix

The grasp matrix $G \in \mathbb{R}^{m \times kp}$ relates the contact forces applied by the manipulators to the net wrench on the object. Here, m is the task-space dimension ($m = 3$ for planar grasps and $m = 6$ for spatial grasps), k is the number of contact points, and p is the number of independent force/moment components per contact. The value of p depends on the assumed contact model (Rimon and Burdick (2019)): $p = 3$ for point contact with friction, $p = 4$ for soft point contact, and $p = 6$ for full contact models with force and moment transmission.

For each contact point C_i , we define a local contact frame with orthonormal vectors $\mathbf{s}_i$, $\mathbf{t}_i$, and $\mathbf{n}_i$, where $\mathbf{n}_i$ is the normal

direction and $\mathbf{s}_i, \mathbf{t}_i$ span the tangent plane. Under a 2D model, only tangential f_t^i and normal f_n^i forces are relevant; in 3D, additional components such as a second tangential force f_s^i and/or moments (e.g., τ_n^i) may be included depending on the contact model.

The grasp matrix for the i^{th} contact point is defined as:

$$G_i(q) = \begin{bmatrix} \mathbf{s}_i & \mathbf{t}_i & \mathbf{n}_i & \mathbf{0}_{3 \times (p-3)} \\ H_i \times \mathbf{s}_i & H_i \times \mathbf{t}_i & H_i \times \mathbf{n}_i & W_i \end{bmatrix},$$

where $H_i = R(\theta) \mathbf{b}_i$. Here, q denotes the object configuration, $\mathbf{b}_i$ is the position vector of contact point C_i expressed in the object frame F_B , and $R(\theta)$ is the rotation matrix from F_B to the world frame F_W .

The submatrix W_i accounts for the additional terms in the wrench mapping when $p > 3$. It is constructed using the same local contact frame basis vectors $\mathbf{s}_i$, $\mathbf{t}_i$, and $\mathbf{n}_i$, but includes only the subset required by the assumed contact model. For soft point contact, W_i typically includes an additional moment term aligned with $\mathbf{n}_i$; for full contact models, it includes all three directions to represent transmissible moments about $\mathbf{s}_i$, $\mathbf{t}_i$, and $\mathbf{n}_i$.

The complete grasp matrix is constructed by horizontally concatenating the individual contact matrices:

$$G(q) = [G_1(q) \quad G_2(q) \quad \dots \quad G_k(q)].$$

This formulation naturally extends to cooperative manipulation scenarios involving dual-arm or multi-fingered systems by treating each manipulator's contact as an independent contribution to the overall wrench.

2.1.2 Hand Jacobian

The hand Jacobian $J_H \in \mathbb{R}^{kp \times N}$ relates the joint velocities of the manipulators/fingers to the velocity of the object's respective contact frame velocities. Here, N denotes the total number of joints across the manipulators in contact.

For each contact point i , the local manipulator or finger Jacobian $J_i(\phi_i) \in \mathbb{R}^{6 \times n_i}$ is transformed into the contact frame using the rotation matrix R_{PK_i} (from the manipulator base frame F_R to the world frame F_W) and the local contact frame basis W_{p,k_i} . In the general case where the contact model includes both force and moment transmission (e.g., full contact,

$p = 6$), this transformation is performed using block-diagonal matrices:

$$W_{\text{block}}^{(i)} = \begin{bmatrix} W_{p,ki}^T & \mathbf{0} \\ \mathbf{0} & W_{p,ki}^T \end{bmatrix}, \quad R_{\text{block}}^{(i)} = \begin{bmatrix} R_{PK_i} & \mathbf{0} \\ \mathbf{0} & R_{PK_i} \end{bmatrix}.$$

The transformed Jacobian for the i^{th} contact is then

$$J_H^{(i)} = W_{\text{block}}^{(i)} R_{\text{block}}^{(i)} J_i(\phi_i).$$

However, in cases with simpler contact models (e.g., point contact with friction where $p = 3$), only the translational components of the Jacobian are relevant, and the transformation simplifies to:

$$J_H^{(i)} = W_{p,ki}^T R_{PK_i} J_i^{\text{trans}}(\phi_i),$$

where $J_i^{\text{trans}} \in \mathbb{R}^{3 \times n_i}$ denotes the top three rows (translational part) of the full Jacobian.

The overall hand Jacobian is assembled in a block-diagonal fashion as:

$$J_H(\phi, q) = \begin{bmatrix} J_H^{(1)} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & J_H^{(2)} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & J_H^{(k)} \end{bmatrix}.$$

This formulation ensures modularity and scalability in multi-arm or multi-fingered systems, allowing each manipulator's Jacobian to be independently transformed and integrated based on the contact model.

2.2 Redundancy

With the grasp matrix $G \in \mathbb{R}^{m \times kp}$ relating contact forces to object wrenches, and the hand Jacobian $J_H \in \mathbb{R}^{kp \times N}$ mapping joint velocities to contact point velocities, we can now formulate a unified kinematic relationship between the object's Cartesian velocity $\dot{q} \in \mathbb{R}^m$ and the robot's joint velocities $\dot{\phi} \in \mathbb{R}^N$. This coupling is central to cooperative manipulation, where multiple arms or fingers interact with a common object. The combined formulation is expressed as

$$J_H \dot{\phi} = G^T \dot{q}. \quad (1)$$

By rearranging this equation, we can express the object's velocity in terms of the joint velocities,

$$\dot{q} = A \dot{\phi}, \quad (2)$$

where A denotes the local object-velocity map induced by the grasp compatibility constraint. In implementation, this map is evaluated using the Moore–Penrose pseudoinverse or a regularized least-squares form, rather than an explicit inverse of G^T , since the grasp matrix is not necessarily square.

The matrix A effectively relates the object's velocity $\dot{q}$ to the robot's joint velocities $\dot{\phi}$. Therefore, the expression for joint velocities becomes

$$\dot{\phi} = A^\dagger \dot{q} + (I_N - A^\dagger A) \dot{\phi}_{\text{opt}}, \quad (3)$$

where $A^\dagger$ denotes the Moore–Penrose pseudoinverse of A . The term $(I_N - A^\dagger A)$ represents the null-space projection matrix of A , while $\dot{\phi}_{\text{opt}}$ is the null-space contribution vector. This null-space contribution term allows for changes in the configuration of the manipulators or robotic hand without altering the position or orientation of the object being held. By exploiting this redundancy, additional secondary objectives can be optimized based on task requirements without affecting the primary task objective.

2.3 Optimization Framework

The secondary joint-velocity term is generated from the gradient of a scalar configuration objective $W(\phi)$ as

$$\dot{\phi}_{\text{opt}} = \Lambda \left(\frac{\partial W}{\partial \phi} \right)^T, \quad (4)$$

where Λ is the optimization gain. A positive value of Λ produces gradient ascent and is used to maximize the objective, whereas a negative value produces gradient descent and is used to minimize it. When projected into the null space of the primary task, this term modifies the redundant joint configuration according to the selected objective without contributing to the commanded object motion.

Most real-world systems operate in discrete time, necessitating a discrete time control approach. Applying Euler discretization to the above equation yields the following discrete-time update

$$\phi(t + \Delta t) = \phi(t) + A^\dagger \dot{q} \Delta t + (I_N - A^\dagger A) \dot{\phi}_{\text{opt}} \Delta t. \quad (5)$$

However, a key drawback of this Euler discretization method is the presence of drift in the system, which accumulates over time and leads to significant errors in tracking the desired trajectory of the object. To mitigate this issue, we introduce a feedback loop into the update equation. The revised discrete-time update is given by

$$\phi(t + \Delta t) = \phi(t) + \underbrace{A^\dagger (\dot{q}_d + K_p e) \Delta t}_{\text{Trajectory tracking term}} + \underbrace{(I_N - A^\dagger A) \dot{\phi}_{\text{opt}} \Delta t}_{\text{Null-space optimization term}}, \quad (6)$$

where $K_p \in \mathbb{R}^{m \times m}$ is a positive definite gain matrix, and the error term e is defined as

$$e = q_d - q_{\text{actual}}. \quad (7)$$

The actual position and orientation of the object, q_{actual} , can be obtained through various methods, such as camera-based tracking or estimation based on the robot's configuration. This feedback loop helps to reduce drift and improve the trajectory tracking performance.

Given that the null-space of the hand Jacobian J_H is a subset of the null-space of the matrix A , we can further refine the approach to redundancy optimization for robot configurations. To exploit this relationship, we modify the control law in equation 6 to better handle the redundant degrees of freedom.

The updated discrete-time equation is

$$\phi(t + \Delta t) = \phi(t) + \underbrace{A^\dagger(q_d + K_p e) \Delta t}_{\text{Trajectory tracking term}} + \underbrace{(I_N - J_H^\dagger J_H) \dot{\phi}_{\text{opt}} \Delta t}_{\text{Null-space optimization term}}, \quad (8)$$

The manipulability metrics do not account for collisions. No additional collision term was required in the planar setup because the motion was constrained and the links were short. In the spatial simulations, a standard potential-based collision penalty was added to the null-space objective and applied consistently across all optimized modes Nakamura (1991); Khatib (1986). This term discourages inter-arm collisions and unintended contact between the manipulators and the object during posture reconfiguration.

2.4 Objective Functions

In this section, we outline the three objective functions considered in this work: velocity manipulability, force manipulability, and directional-force manipulability.

The contact compatibility constraint in object grasping and cooperative manipulation tasks is given by

$$J_H(\phi, q) \dot{\phi} = G^T(q) \dot{q}. \quad (9)$$

This relation couples the joint velocities of the manipulators to the object velocity through the hand Jacobian and the grasp matrix. In implementation, the corresponding local object-velocity map is evaluated in a least-squares sense as

$$\dot{q} = A(\phi, q) \dot{\phi}, \quad (10)$$

where $A(\phi, q)$ denotes the local map from joint velocity to object velocity induced by the grasp compatibility constraint. Since the grasp matrix is not necessarily square, $A(\phi, q)$ is computed using the Moore–Penrose pseudoinverse or a regularized least-squares form, rather than by explicitly inverting $G^T(q)$.

For a unit-bounded joint velocity input, we impose

$$\dot{\phi}^T \dot{\phi} \leq 1. \quad (11)$$

Using the implemented object-velocity map in equation (10), the corresponding velocity manipulability ellipsoid is evaluated as

$$\dot{q}^T (AA^T)^\dagger \dot{q} \leq 1. \quad (12)$$

This expression is used as the numerical manipulability model in the controller.

2.4.1 Velocity Manipulability

From equation (12), AA^T is the velocity-capability matrix, with scalar manipulability index

$$W_v = \sqrt{\det(AA^T)}. \quad (13)$$

Larger values indicate greater overall task-space velocity capability.

2.4.2 Force Manipulability

By velocity–force duality, $(AA^T)^\dagger$ is the force-capability matrix, equal to $(AA^T)^{-1}$ for full-row-rank A , with scalar index

$$W_f = \sqrt{\det((AA^T)^\dagger)}. \quad (14)$$

The velocity- and force-capability ellipsoids share principal directions with reciprocal axis lengths; hence, maximum force corresponds to minimum velocity capability Nakamura (1991).

2.4.3 Directional Force Manipulability

The scalar force-manipulability measure in equation (14) characterizes the overall size of the force-capability ellipsoid. In many cooperative manipulation tasks, however, the relevant force requirement is direction dependent because the commanded object motion and the dominant external load may act along different directions. This situation arises, for example, when the object is moved while resisting gravity, pushed along a surface while compensating for a lateral disturbance, or guided along a tool or insertion direction while opposing a load in another direction. The desired load direction is represented by the positive-semidefinite weighting matrix F , whose nonzero entries identify the relevant force and moment components. In such cases, increasing the overall force-manipulability volume does not necessarily improve the robot's ability to generate or resist force along the task-relevant direction.

To account for this directional requirement, we consider two complementary directional-force configuration-shaping objectives. The prescribed load direction is encoded by the positive-semidefinite weighting matrix

$$F = \text{diag}(|F_x|, |F_y|, |F_z|, |M_x|, |M_y|, |M_z|), \quad (15)$$

where nonzero entries identify the task-relevant force and moment components. In the reported evaluations, only one force component is active, so $F / \text{tr}(F)$ reduces to a one-hot selector of the prescribed load direction; all moment entries are zero. When force and moment components are used together, consistently scaled task-space coordinates are assumed. Since the objective compares directional structure rather than magnitude, both matrices are trace-normalized. The first formulation is a *direct force-space objective*, defined as the Frobenius distance between the normalized force-capability matrix $(AA^T)^\dagger$ and the normalized weighting matrix F :

$$W_{df}^{\text{direct}} = \sqrt{\text{tr} \left[\left(\frac{(AA^T)^\dagger}{\text{tr}((AA^T)^\dagger)} - \frac{F}{\text{tr}(F)} \right) \left(\frac{(AA^T)^\dagger}{\text{tr}((AA^T)^\dagger)} - \frac{F}{\text{tr}(F)} \right)^T \right]}. \quad (16)$$

The direct formulation is minimized. Accordingly, minimizing W_{df}^{direct} encourages the normalized force-capability

structure to approach the prescribed directional structure represented by F .

We additionally consider an *indirect velocity-dual objective*, following the redundancy-based force-interaction formulation in [Jadav and Palanthandalam-Madapusi \(2024\)](#). This objective compares the normalized velocity-capability matrix with the same prescribed directional matrix:

$$W_{df}^{\text{indirect}} = \sqrt{\text{tr} \left[\left(\frac{AA^T}{\text{tr}(AA^T)} - \frac{F}{\text{tr}(F)} \right) \left(\frac{AA^T}{\text{tr}(AA^T)} - \frac{F}{\text{tr}(F)} \right)^T \right]}. \quad (17)$$

The indirect formulation is maximized. Maximizing W_{df}^{indirect} drives the normalized velocity-capability structure away from the prescribed load direction and thereby shapes the configuration indirectly through velocity-force duality.

Both formulations allow the load direction to be prescribed independently of the commanded object motion and are therefore applicable when the motion and external-load directions differ. However, the direct formulation operates in force-capability space, whereas the indirect formulation operates in velocity-capability space. They consequently share the same task-level motivation but are not generally mathematically equivalent.

The planar hardware experiments employ the indirect velocity-dual formulation in equation (17), whereas the spatial directional-force simulations employ the direct force-space formulation in equation (16). Both formulations are additionally evaluated under matched conditions in the static spatial comparison. The force-space derivation, the effect of trace normalization, and the interpretation of the indirect formulation are provided in Appendix A.

3. Experimental Evaluation: Planar Dual-Arm Hardware

3.1 Experimental Setup

Figure 3 shows the planar dual-arm platform on which all experiments are conducted. The rig comprises two identical four-revolute (4R) manipulators, each built from Dynamixel actuators driven through a U2D2 interface and interconnected

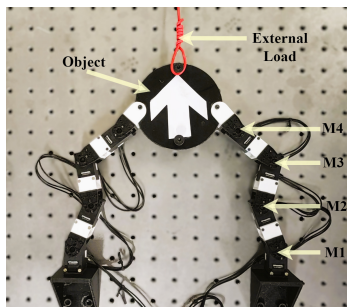
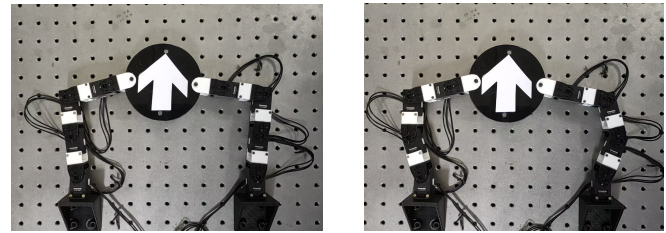
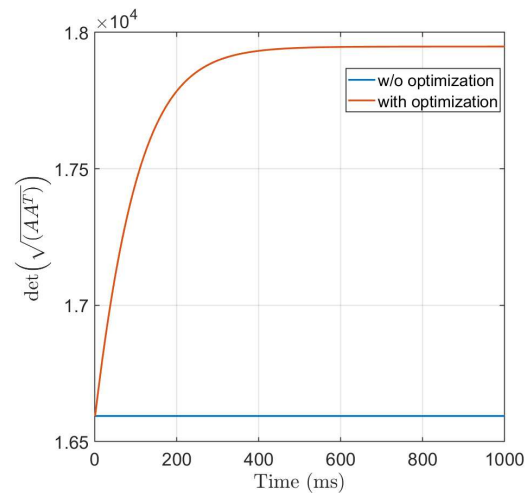


Figure 3: Planar dual-arm hardware setup with two 4R manipulators pin-jointed to a circular object under a suspended directional load. **Source:** Authors' own work.



(a) Without optimization

(b) With optimization



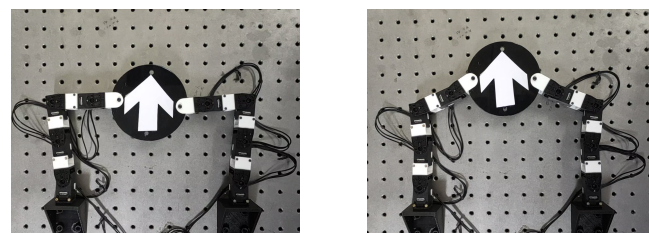
(c) Velocity manipulability index vs time

Figure 4: Static object holding with and without null-space redundancy optimization for velocity manipulability. **Source:** Authors' own work.

by lightweight, 3-D-printed links. Both wrists grasp a circular plate via pin joints, ensuring that the relative motion between the end-effectors and the object is fully constrained and purely determined by the arm kinematics. This rigid coupling is crucial since the proposed controller operates at the kinematic level, without directly modeling contact forces or using torque feedback. The pin-jointed grasp enforces a closed kinematic chain, allowing the object pose to be reliably controlled through joint space coordination alone. A high-contrast arrow printed on the plate serves as a clear fiducial for orientation tracking. Provisions on the plate allow a weight to be attached, supplying a steady force in the required direction during the force and directional-force manipulability trials.

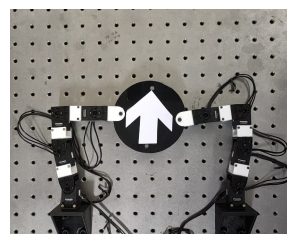
3.2 Velocity-Manipulability Experiments

To illustrate how the proposed redundancy optimization law (equation 8) improves velocity manipulability, we designed three planar dual-arm experiments that progressively increase task complexity: (i) static holding, (ii) pure translational motion, and (iii) in-plane rotation. In every case, the experiment is executed twice, first with the null-space optimization term of equation 8 disabled and then with it enabled. Through-

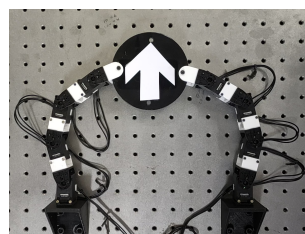


(a) Without optimization: Start

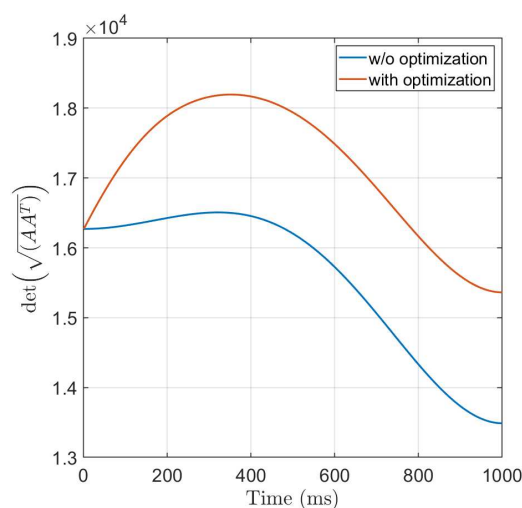
(b) Without optimization: End



(c) With optimization: Start



(d) With optimization: End



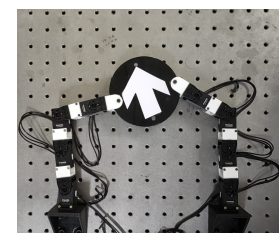
(e) Velocity manipulability index vs time

Figure 5: Moving the object in the Y-direction with and without redundancy optimization for the velocity manipulability. **Source:** Authors' own work.

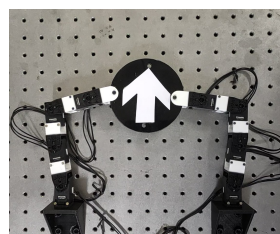
out the hardware and simulation results, the terms *baseline*, *no optimization*, and *without optimization* denote the same control mode, in which the primary object-tracking controller remains active while the manipulability-based null-space term is disabled. For each comparison, the baseline metric is evaluated along the joint-state trajectory generated in this mode, whereas the optimized metric is evaluated along the trajectory generated with the corresponding objective active. The velocity-manipulability metric in equation 13 is monitored throughout these trials, while Figures 4–6 illustrate the corresponding posture evolution and allow qualitative inspection



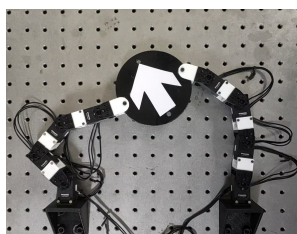
(a) Without optimization: Start



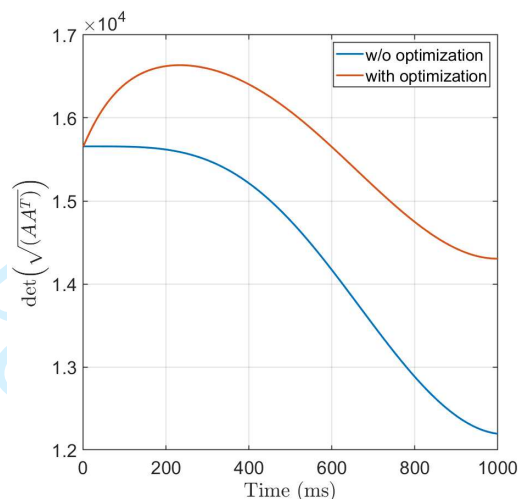
(b) Without optimization: End



(c) With optimization: Start



(d) With optimization: End



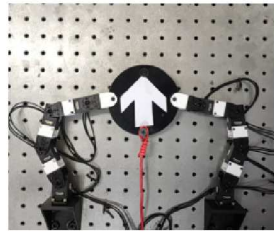
(e) Velocity manipulability index vs time

Figure 6: Rotating the object with and without redundancy optimization, using the velocity manipulability cost function. **Source:** Authors' own work.

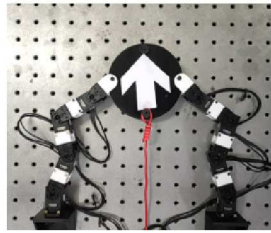
of the preservation of the commanded object motion. Quantitative object-tracking performance for the spatial evaluation is reported separately in Section 4 and Table I.

3.2.1 Static Holding

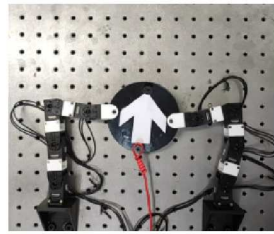
Starting from an arbitrary, feasible grasp position, both arms are commanded to maintain the object pose ($\dot{\mathbf{q}} = \mathbf{0}$). When redundancy optimization is inactive, the arms freeze in their initial configuration (Figure 4a), and the manipulability remains constant. When it is active, the null-space gradient gradually steers the elbows apart (Figure 4b), yielding a rise



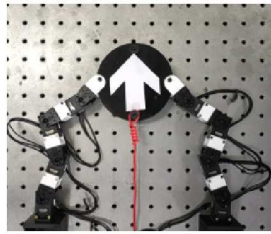
(a) Force Manipulability: Start



(b) Force Manipulability: End



(c) Directional Force: Start



(d) Directional Force: End

Figure 7: Comparison of joint configurations produced by force-manipulability and indirect directional force manipulability optimization during motion in the positive y-direction with an external force applied in the negative y-direction. **Source:** Authors' own work.



(a) Directional Force: Start



(b) Directional Force: End

Figure 8: Joint configuration under redundancy optimization using the indirect directional force manipulability cost function, with motion in the positive y-direction and force applied in the positive x-direction. **Source:** Authors' own work.

in the velocity manipulability index (Figure 4c) while the object pose holds its initial position.

3.2.2 Pure Y-Direction Translation

The second test commands a translation along the positive y-axis of the world frame. Without redundancy optimization (Figures 5a–5b), the robot successfully tracks the trajectory but exhibits lower velocity manipulability (Figure 5c). When optimization is activated (Figures 5c–5d), the null-space gradient redistributes the joint angles (Figure 5d), resulting in the preservation of a higher manipulability index (solid red curve in Figure 5e).

3.2.3 In-Plane Rotation About the Object Centre

Finally, the object is commanded to rotate about its own z-axis. Without redundancy optimization (Figures 6a–6b), the robot fails to complete the full desired motion and exhibits low velocity manipulability throughout (Figure 6e). When the optimization is enabled, the joints are steered into more favorable configurations (Figure 6d), allowing the robot to complete the motion while maintaining a significantly higher velocity manipulability index (Figure 6e).

3.3 Force and Directional-Force Manipulability (Experiments)

In many manipulation tasks, the ability to exert or resist forces is as critical as achieving velocity dexterity. Effective force manipulability enables more balanced load distribution across joints, reducing the likelihood of actuator saturation and minimizing motor stress during high-load operations. To evaluate the benefits of force-based redundancy optimization, we repeat the y-direction translation experiment from Section 3.2 under a constant external load and monitor both manipulability indices and overall control effort. The control effort is approximated using joint current measurements from the Dynamixel motors via the metric $\sqrt{ii^T}$, which serves as a proxy for actuator workload and is analogous to torque-based effort in traditional manipulators.

Two experiments are performed. First, we compare three null-space objectives during motion with a load in the negative y-direction: (i) no optimization (baseline), (ii) force manipulability optimization (equation 14), and (iii) directional force manipulability optimization (equation 17). Second, to highlight the advantage of directional redundancy, we repeat the task with the external load applied along the positive x-direction while motion remains along the positive y-axis. Initial and final joint configurations are shown in Figures 7 and 8; a qualitative summary appears in Figure 9.

3.3.1 Translation in Positive Y with Constant Force in –y Direction

As illustrated in Figure 7, the joint configurations under force-based redundancy optimization differ significantly from those during velocity manipulability trials (Figure 5). In contrast to velocity optimization—which encouraged the arms to spread apart—force-based strategies cause the manipulators to move closer together and shift toward the structural side of the setup, as seen in Figures 7b and 7d.

Quantitatively, Figure 9a shows that the force manipulability index improves consistently when redundancy optimization is enabled, outperforming the baseline across the trajectory. Directional force manipulability, depicted in Figure 9b, also improves, although the gains are more modest. Initially, both optimized and non-optimized curves follow similar trends, but diverge over time, with the directional-force strategy achieving superior performance at the end.

Figure 9c compares actuator-current demand for the aligned

planar hardware condition. The baseline shows the highest current-vector norm over most of the trajectory, force-manipulability optimization the lowest, and Directional-Force Manipulability (Indirect) an intermediate response. This result is specific to this condition and does not imply that null-space optimization generally minimizes overall effort.

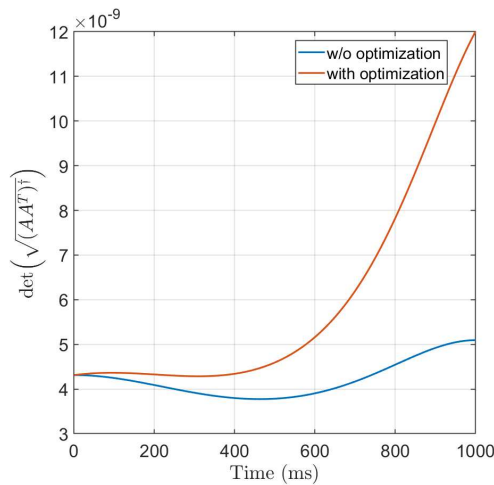
3.3.2 Translation in Positive Y with Constant Force in x Direction

To further highlight the importance of directional-force manipulability, we consider a task in which the motion and external load act in different directions—a common situation in many

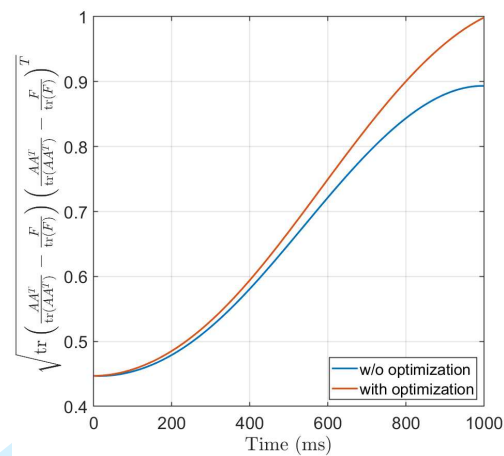
manipulation scenarios. Specifically, the robot is commanded to translate along the positive y-axis while resisting an external force applied in the positive x-direction. Figure 8 illustrates the resulting postures. The control effort profile in Figure 9d reveals that force-only redundancy now incurs a higher effort cost, whereas directional-force optimization produces lower actuator-current demand than force-manipulability optimization under the evaluated non-aligned motion/load condition.

4. Experimental Evaluation: Spatial Dual-Arm Simulation

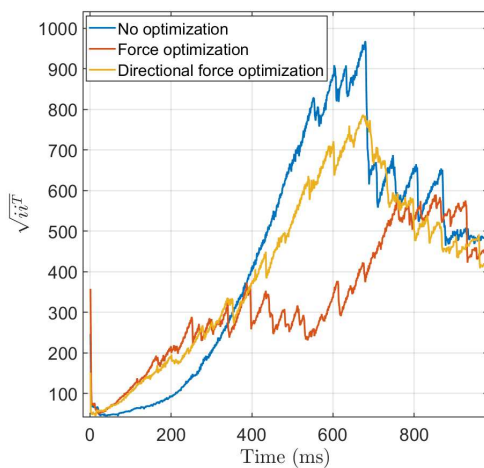
To assess the framework in spatial, high-degree-of-freedom cooperative manipulation, it was implemented in MuJoCo



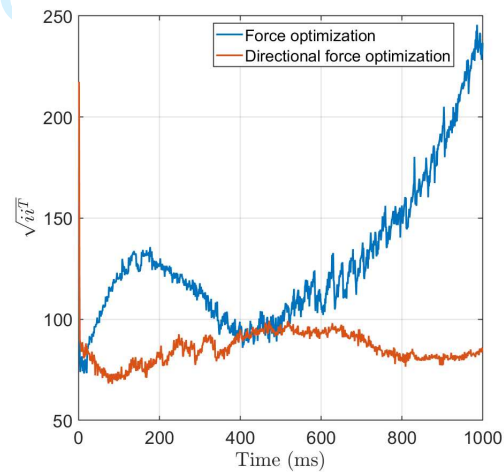
(a) Force manipulability index



(b) Directional-force manipulability (Indirect)



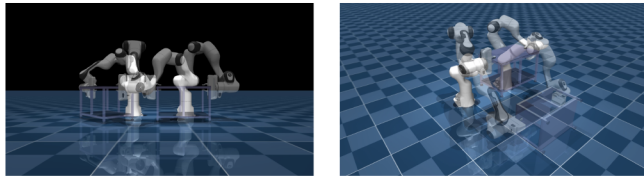
(c) Effort for aligned motion and load



(d) Effort for non-aligned motion and load

Figure 9: Comparison of force manipulability and directional-force manipulability (indirect). The first row compares force and directional-force manipulability (indirect) indices during y-direction motion with a $-y$ external load. The second row compares actuator-current effort, measured using $\sqrt{ii^T}$. In (c), the commanded motion and external load are along the same task direction. In (d), the object is commanded to move along the y-direction while the external force is applied along the x-direction, representing a non-aligned motion/load condition. Directional-force manipulability is especially useful in the non-aligned case because it explicitly biases the redundant configuration toward the task-relevant force direction. **Source:** Authors' own work.

using two 7-DoF Franka Panda manipulators. The robots manipulate a freely moving rigid object through simulated gripper-object contact, with the object pose fed back to the closed-loop inverse-kinematics controller in equation 8.



(a) Pose 1: front view (b) Pose 4: perspective view

Figure 11: Superimposed snapshots of representative six-dimensional trajectories. **Source:** Authors' own work.

Three manipulability-based optimization modes are evaluated against a baseline with null-space optimization disabled: (i) velocity-manipulability optimization, (ii) force-manipulability optimization, and (iii) directional-force optimization. The direct directional-force formulation is used in the primary comparisons, while the indirect formulation is additionally evaluated in the static and six-dimensional cases. In all cases, the object is first grasped using Mink-based differential inverse kinematics Zakka (2026). The reported comparison begins only after the grasp has been established, at which point the corresponding null-space optimization mode

is activated. The comparison considers three task scenarios: (a) static holding at a prescribed object pose, (b) vertical pick-and-place, in which the object is lifted to a prescribed height and returned to its initial pose without commanded rotation, and (c) spatial pick-and-place, in which the object follows a six-dimensional trajectory combining translational and rotational motion. Each scenario is evaluated from six initial object configurations to assess sensitivity to the starting pose.

4.1 Static Redundancy Optimization

Figure 10 demonstrates consistent objective-specific behavior across the six static initial configurations. Relative to its matched baseline, velocity and force manipulability increase, whereas the directional-force distance decreases, while the prescribed object pose remains unchanged. The magnitude of improvement and the time required for each metric to approach a steady value vary across both the initial configurations and the selected objectives, as reflected by the different curve separations and time spans in the figure. This variation is expected because the initial posture determines the available null-space directions, the local objective gradient, and the extent of feasible joint reconfiguration. Accordingly, the principal result is the consistent direction of improvement for each objective, rather than an identical gain or convergence rate across all configurations. The posture-level effect of the different redundancy objectives is illustrated in Figure 12. While

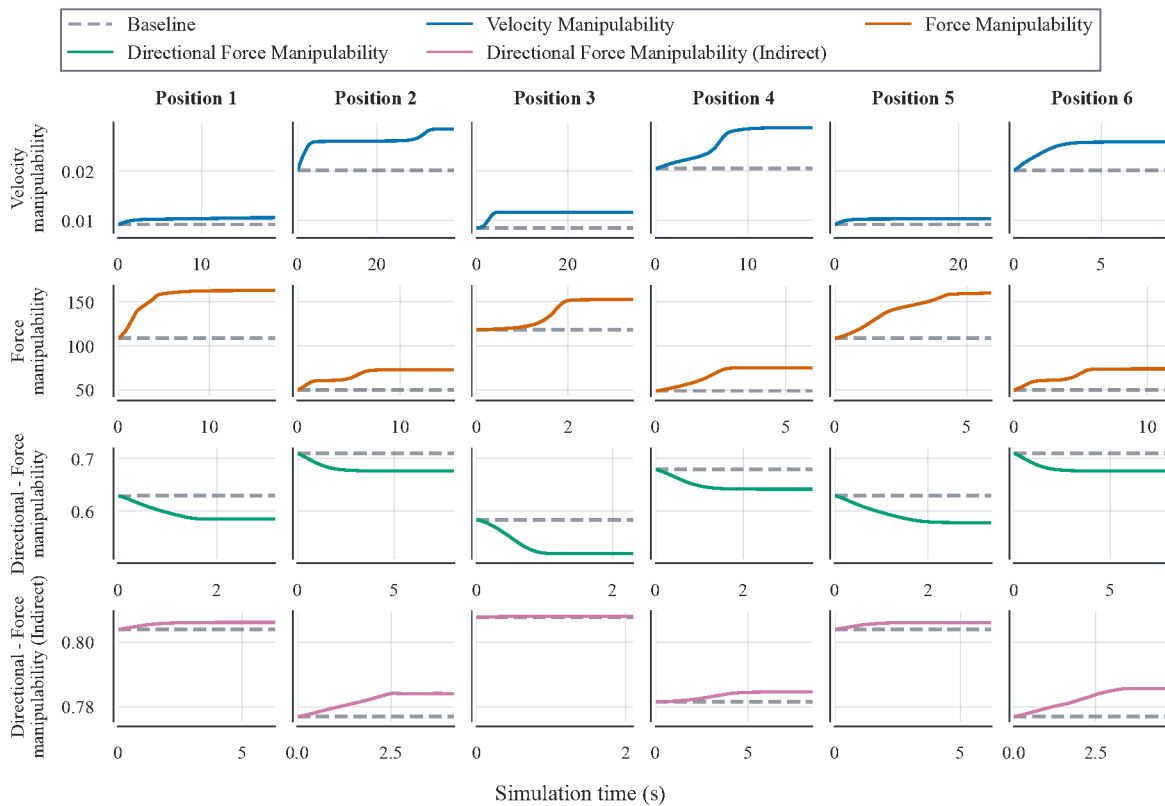


Figure 10: Static redundancy-optimization results for six initial object configurations. The rows show velocity manipulability, force manipulability, and the direct and indirect formulations of directional-force manipulability. **Source:** Authors' own work.

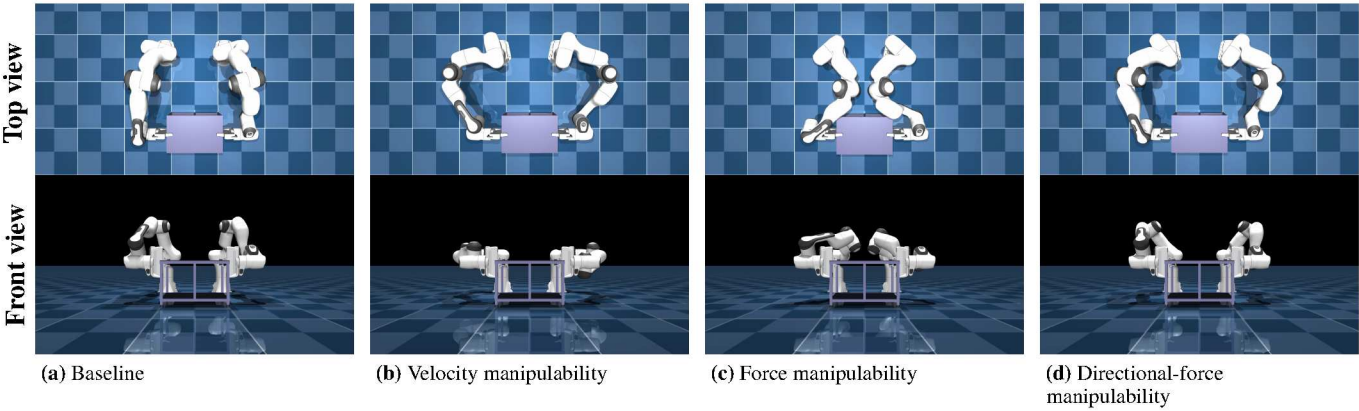


Figure 12: Final dual-arm configurations during static holding under (a) the baseline, (b) velocity-manipulability optimization, (c) force-manipulability optimization, and (d) directional-force-manipulability optimization. In all cases, the prescribed object pose is maintained while the redundant arm configuration changes. **Source:** Authors' own work.

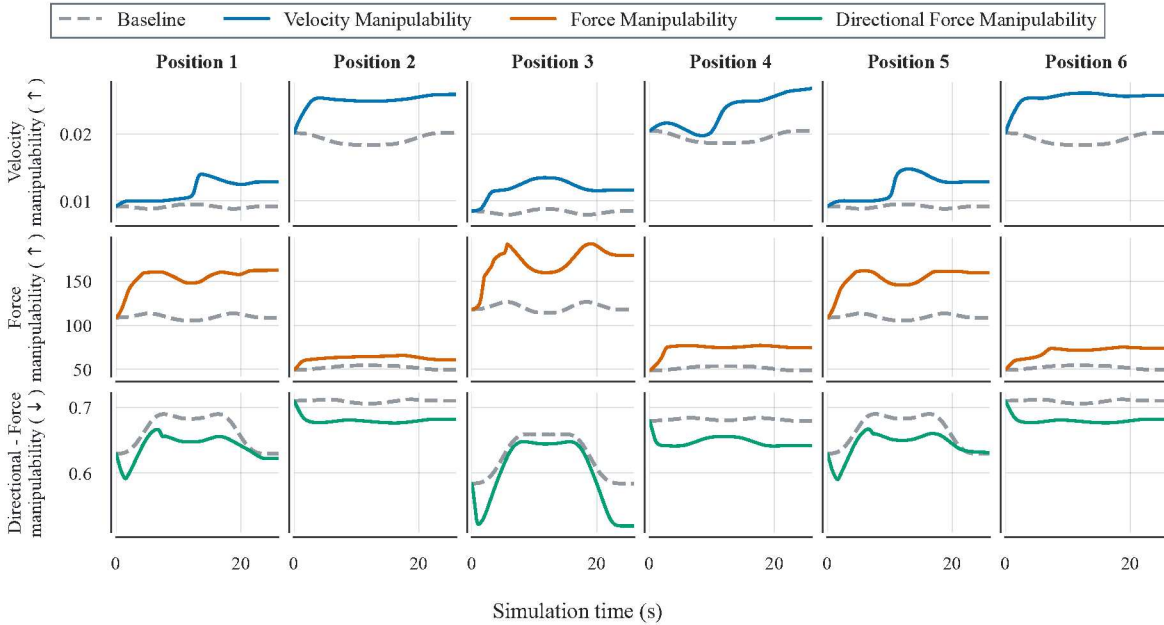


Figure 13: Redundancy-optimization results for translational pick-and-place from six initial object positions. Higher velocity and force manipulability values are preferred, while lower directional-force distance indicates better agreement with the prescribed wrench direction. **Source:** Authors' own work.

the object is maintained at the same prescribed static pose, the baseline, velocity-manipulability, force-manipulability, and directional force manipulability modes produce visibly different internal dual-arm configurations. The differences in elbow placement, arm spread, and joint arrangement demonstrate that each objective exploits the available kinematic redundancy differently.

4.2 Translational Pick-and-Place

During translational pick-and-place, the commanded object motion continuously changes the arm configuration, causing both the baseline and optimized metrics to vary over time,

as shown in Figure 13. Despite this trajectory-induced variation, each objective-specific controller generally shifts its corresponding metric in the desired direction relative to the matched baseline: velocity and force manipulability remain higher, whereas the directional-force distance remains lower. For the directional-force objective, a lower value indicates better alignment of the force-capability structure with the prescribed wrench direction because this metric is minimized. The magnitude and temporal profile of the improvement depend on both the initial configuration and the current trajectory segment. Velocity gains are particularly pronounced at Positions 2, 4, and 6, whereas the separation from the baseline for

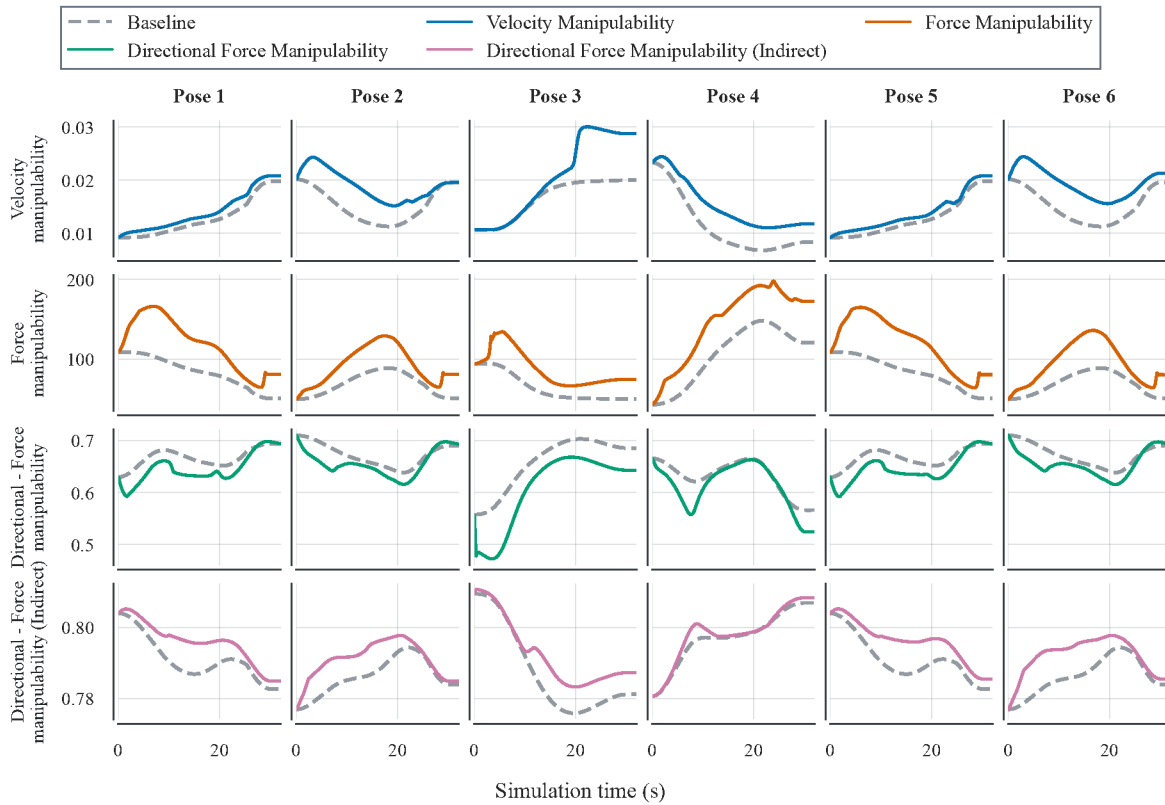


Figure 14: Six-dimensional pick-and-place results across six initial object poses for the baseline and three manipulability objectives, with the directional-force objective evaluated using its direct and indirect formulations. **Source:** Authors' own work.

the force-based objectives is comparatively modest in some cases, especially at Positions 2 and 6. As in the static case, the improvement is configuration dependent; however, during translational motion, the primary task also continuously changes the available null-space directions, the local objective gradient, and the extent of feasible posture reconfiguration along the trajectory.

4.3 Six-Dimensional Pick-and-Place

Figure 14 shows that the objective-specific trends observed in the translational task are retained during simultaneous translation and rotation. Across the six initial poses, velocity and force optimization generally maintain their respective metrics above the matched baselines, whereas the direct directional-force objective maintains a lower distance from the prescribed wrench direction. The improvement varies with the initial pose and trajectory phase because the evolving task constraints change the null-space motion available for posture reconfiguration.

4.4 Comparison of Direct and Indirect Directional-Force Objectives

The direct and indirect objectives, defined in Section 2.4.3 and derived in Appendix A, are compared under static-holding and six-dimensional pick-and-place conditions. Static holding isolates local posture adaptation at a fixed object pose, whereas

the six-dimensional task evaluates the trajectory-dependent consequences of the two objectives. The translational task was not duplicated because the six-dimensional case provides the more demanding moving comparison. Since the two objective values are not directly comparable, the analysis focuses on actuator effort and the resulting joint configurations.

The six-dimensional trajectories comprise two motion patterns, represented by Poses 1 and 4 in Figure 11. Poses 1, 2, 5, and 6 follow oblique lift-transfer-place motions combining translation in the x -, y -, and z -directions with rotation about the z -axis. Poses 3 and 4 remain in the x - z plane and include an out-of-plane reorientation during lifting, producing more strongly gravity-coupled motions. These patterns provide spatial analogues of the aligned and non-aligned hardware cases: planar motions are more gravity-aligned, whereas oblique motions introduce components not aligned with gravity.

Figure 15 shows that the indirect objective requires less integrated actuator effort than force-manipulability optimization in all six poses. The direct objective does so in five poses, with Pose 3 as the exception. For Poses 3 and 4, the commanded motion is more strongly coupled with the gravity-dominated load direction, reducing the relative benefit of directional posture shaping. Their different outcomes nevertheless show that the initial configuration and resulting posture trajectory also influence the effort. The baseline

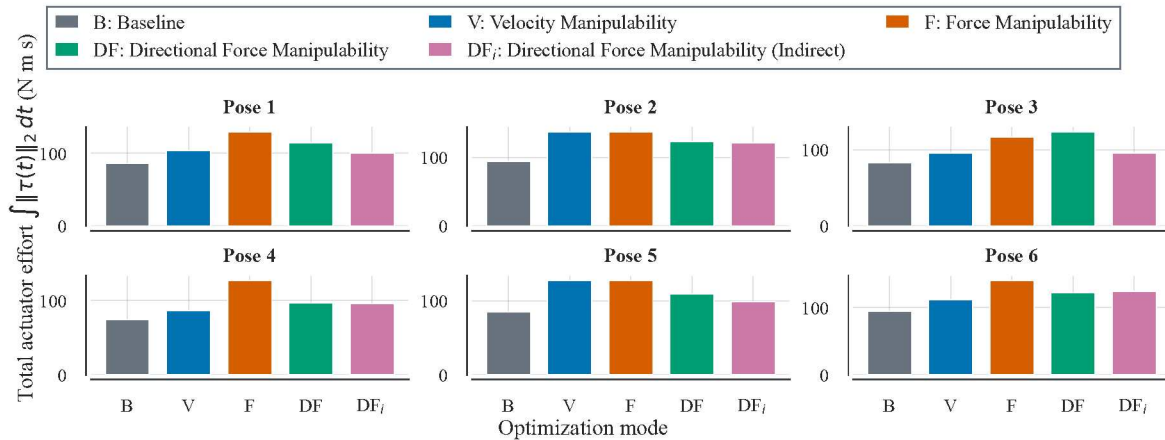


Figure 15: Total actuator effort, $\int \|\tau(t)\|_2 dt$, during six-dimensional pick-and-place for the baseline (B), velocity (V), force (F), direct directional-force (DF), and indirect directional-force (DF_i) objectives. **Source:** Authors' own work.

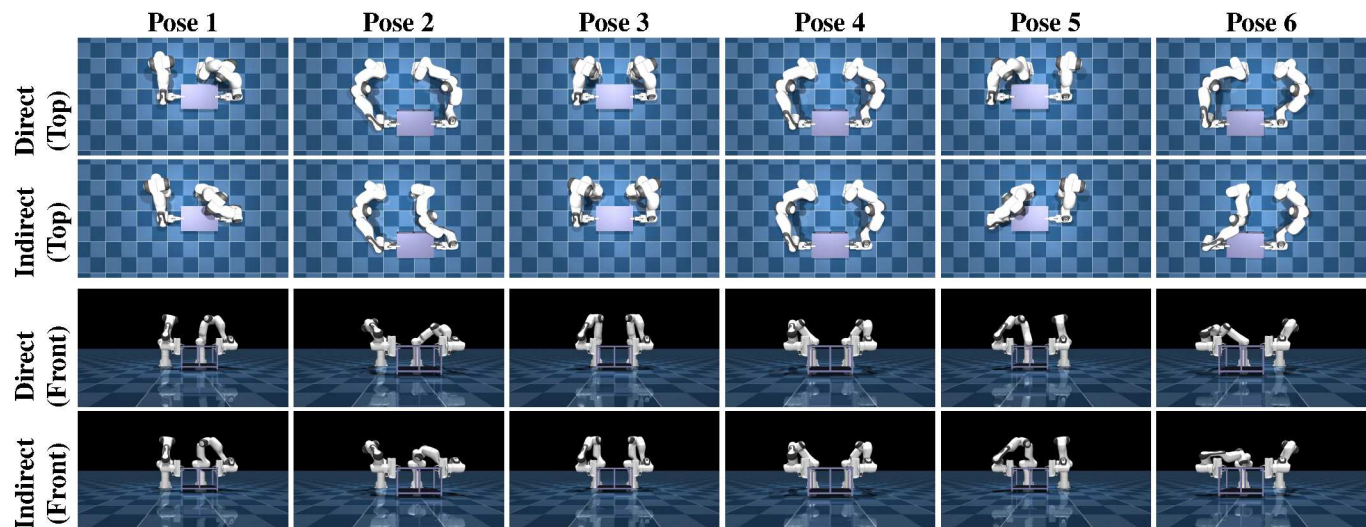


Figure 16: Final configurations produced by the direct and indirect directional-force objectives from six initial object poses, shown from the top and front views. **Source:** Authors' own work.

remains lowest because it introduces no secondary posture-reconfiguration motion; thus, the comparison characterizes the torque-demand consequences of posture shaping rather than direct energy minimization.

Figure 16 compares the static final configurations obtained under identical object poses and load directions. The differences are pronounced for Poses 2 and 6 and subtler for Poses 1, 3, and 5 when the top and front views are considered together. Only Pose 4 produces nearly identical configurations. The two objectives can therefore induce different local posture adaptations even under the same static task conditions.

4.5 Object-Tracking Accuracy and Computational Cost

Table 1 shows that the secondary objectives reshape the redundant configuration while maintaining the primary object-tracking task. Tracking errors remain low for the static and translational cases and increase for the more demanding six-

dimensional trajectories, where translation and rotation must be followed simultaneously, but remain small overall.

The main computational penalty arises from the finite-difference evaluation of the manipulability gradient, which increases the 95th-percentile update time from approximately 4 ms for the baseline to 65.8–74.4 ms with optimization. The underlying closed-loop controller requires only approximately 1.5–1.7 ms, indicating that more efficient gradient evaluation is the principal requirement for real-time deployment.

5. Conclusion

This work compared velocity, force, and directional-force manipulability objectives within a common null-space redundancy-resolution framework for cooperative manipulation. The planar hardware experiments showed that these objectives produce distinct redundant configurations while retaining the commanded object motion. The directional-force

Table I: Object-tracking accuracy and per-step computation latency over six initial configurations. Optimized entries show the range across velocity, force, and directional-force optimization. **Source:** Authors' own work.

Task	Position RMSE baseline (mm)	Position RMSE optimized (mm)	Orientation RMSE baseline (deg)	Orientation RMSE optimized (deg)	Compute time, p95 baseline (ms)	Compute time, p95 optimized (ms)
Static	0.05	0.68–2.07	0.01	0.12–0.34	4.0	67.5–74.4
Translational	0.62	0.93–3.30	0.03	0.10–0.97	4.0	66.4–72.0
pick-and-place						
6D pick-and-place	4.51	4.78–11.14	1.88	1.98–2.25	4.7	65.8–70.7

objective reduced actuator-current effort relative to force-manipulability optimization when the commanded motion and external-load directions were not aligned.

The dual-Franka simulations extended the comparison to static, translational, and six-dimensional manipulation from multiple initial configurations. Each objective generally shifted its corresponding metric in the intended direction while maintaining bounded object-pose tracking errors, although these errors increased during the more demanding six-dimensional motion. The principal implementation limitation is the finite-difference gradient evaluation, which increased the 95th-percentile update time to 65.8–74.4 ms; analytical or automatic-differentiation-based gradients are therefore needed for higher-rate deployment. Overall, the redundancy objective should be selected according to the task-relevant motion and load requirements.

6. Appendices

A. Derivation and Interpretation of the Directional-Force Objectives

A.1 Direct force-space formulation

Starting from the object-velocity map in equation (10),

$$\dot{q} = A\dot{\phi}, \quad (18)$$

let $w \in \mathbb{R}^m$ denote the object wrench and $\tau \in \mathbb{R}^N$ the corresponding generalized joint torque. Equality of instantaneous power gives

$$\begin{aligned} \tau^T \dot{\phi} &= w^T \dot{q} = w^T A \dot{\phi} \\ \implies \tau &= A^T w. \end{aligned} \quad (19)$$

This is the standard differential kinematics–statics duality obtained from virtual work Nakamura (1991). Under the unit joint-torque bound $\tau^T \tau \leq 1$, equation (19) gives

$$w^T A A^T w \leq 1. \quad (20)$$

For a full-row-rank configuration, $(A A^T)^\dagger = (A A^T)^{-1}$; hence, $(A A^T)^\dagger$ defines the shape of the corresponding force-capability ellipsoid. This establishes why the direct objective uses the force-space matrix $(A A^T)^\dagger$, rather than the velocity-capability matrix $A A^T$.

To compare directional structure independently of overall magnitude, the force-capability matrix and the prescribed weighting matrix are normalized by their traces. Their Frobenius distance is

$$W_{df}^{\text{direct}} = \left\| \frac{(A A^T)^\dagger}{\text{tr}((A A^T)^\dagger)} - \frac{F}{\text{tr}(F)} \right\|_F. \quad (21)$$

Using $\|X\|_F = \sqrt{\text{tr}(X X^T)}$, equation (21) directly yields the direct directional-force objective in equation (16). Trace normalization removes the overall scale while retaining the relative directional structure of each matrix. Because the resulting distance measures mismatch between the normalized force-capability and prescribed directional structures, it is minimized using the general gradient update in equation (4) with a negative optimization gain.

A.2 Indirect velocity-dual formulation

The indirect formulation operates on the velocity-capability matrix $A A^T$, following the redundancy-based force-interaction formulation in Jadav and Palanhandalam-Madapusi (2024):

$$W_{df}^{\text{indirect}} = \left\| \frac{A A^T}{\text{tr}(A A^T)} - \frac{F}{\text{tr}(F)} \right\|_F. \quad (22)$$

Using $\|X\|_F = \sqrt{\text{tr}(X X^T)}$ recovers equation (17). Maximizing this distance separates the normalized velocity-capability structure from the prescribed load direction and, through velocity–force duality, indirectly shapes the redundant configuration for that direction. The objective is therefore maximized using equation (4) with a positive gain.

The direct and indirect formulations are not generally equivalent: the former minimizes mismatch in force-capability space, whereas the latter maximizes separation in velocity-capability space. Consequently, they may produce different redundant configurations, as examined in Section 4.4.

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Supplementary Material

Code and video available at: <https://debojit-d.github.io/Bimanual-Redundancy-Optimization/>.

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